

Assignment 1

February 16, 2026

Question 1. (30 points) Please discuss the following statements and briefly explain your opinion on them in a clear and concise way. Keep in mind that these are not true or false statements.

1. If a linear programming problem has a feasible solution, it is guaranteed to have an optimal solution.
2. If the feasible region of a linear programming problem is unbounded, it means the objective function can be improved indefinitely.
3. If an LP is feasible, then it has an optimal solution that is a corner point.
4. The graphical solution method can only be used to solve linear programs with two decision variables.

Correct Answer 1.

1. While a linear programming (LP) problem with a feasible solution means there is at least one point that satisfies all constraints, it does not guarantee the existence of an optimal solution. For example, in an unbounded LP problem, the objective function may increase (in a maximization problem) or decrease (in a minimization problem) without bound, resulting in no finite optimal solution.
2. An unbounded feasible region does not necessarily imply that the objective function can be improved indefinitely. Whether the objective function can be improved depends on the direction of optimization. For instance, in a maximization problem, if the feasible region is unbounded in a direction where the objective function increases, then the objective function can be improved indefinitely. However, in some cases, the unbounded region might not coincide with the direction of improvement of the objective function, meaning that an optimal solution might still exist even in an unbounded region.
3. While linear programming problems often have optimal solutions at corner points (vertices) of the feasible region, this is not always guaranteed. There are exceptions, such as when the LP is unbounded, in which case the objective function may continue to improve indefinitely without ever reaching a finite optimal solution. Therefore, being feasible does not necessarily imply that an optimal solution exists, and if it does, it doesn't always have to occur at a corner point.
4. The graphical solution method is typically used for solving linear programming problems with two decision variables because it allows easy visualization of the feasible region and objective function. However, the method can, in theory, be extended to higher dimensions (three or more variables), but doing so is often impractical because it becomes difficult or impossible to visually represent higher-dimensional feasible regions. In higher dimensions, algebraic or computational methods (such as the Simplex method) are used instead.

Question 2. (15 points) I also would like you to discuss these statements with your classmates and tell me how their opinion differ from yours (if they do). List out the uni's of every student you have discussed the statements with.

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Question 3. (55 points) A shelter dedicated to supporting victims of human trafficking is tasked with procuring essential resources to provide comprehensive care. These resources include both cleaning supplies and personal care items, crucial for maintaining hygiene and health. The shelter procures these items from different suppliers, given in Table 1. Each month, the shelter places one order with each supplier.

Given the limited funding and different costs of resources from each supplier, the shelter must strategically plan its purchases to minimize expenses while ensuring it meets the necessary service standards for effective victim support. The minimum required quantities for these standards are listed in Table 3. The shelter has a \$2,000 monthly budget for all the resources.

Deliverables:

1. Include a clear description of the decision variables, the objective function, and each constraint. (10 points)
2. Provide your algebraic LP formulation. (20 points)
3. Solve the LP using any solver (e.g., Gurobi, CPLEX) and share the code/model you used for solving the problem. A screenshot would also work. (20 points)
4. Explain your solution. Discuss the implications of your solution regarding the shelter's operations and resource allocation strategies. (5 points)

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Item	Cleaning				Personal Care			
Supplier	Toilet Paper (1 roll)	Liquid Soap (lt.)	Detergent (lt.)	Cloth (m ²)	Tooth paste	Tooth brush	Pads	Shampoo (lt.)
1	\$0.80	\$6.40	\$6.80	\$10.00	\$2.60	\$0.80	\$0.20	\$2.30
2	\$0.95	\$3.98	\$4.60	\$11.00	\$3.00	\$0.85	\$0.18	\$1.20
3	\$0.84	\$5.50	\$7.50	\$10.50	\$2.80	\$0.82	\$0.15	\$3.00

Table 1: Cost of acquiring needed resources from different suppliers.

Item	Cleaning				Personal Care			
Supplier	Toilet Paper (rolls)	Liquid Soap (lt.)	Detergent (lt.)	Cloth (m ²)	Tooth paste	Tooth brush	Pads	Shampoo (lt.)
1	150	25	20	10	50	50	150	20
2	100	15	10	10	60	60	100	20
3	70	30	15	15	30	30	100	30

Table 2: Maximum supply capacity of cleaning and personal care items from different suppliers.

Type of Resource	Minimum Required Quantity (units)
Toilet Paper (rolls)	200
Liquid Soap (liters)	40
Detergent (liters)	30
Cloths (square meters)	20
Toothpaste (tubes)	100
Toothbrushes	100
Sanitary Pads (packs)	300
Shampoo (liters)	40

Table 3: Minimum required quantities of cleaning and personal care items for a shelter housing 20 people. Quantities are estimated for one month of supply.

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To solve the problem using a solver, you can use the dataset given below.

```
# Data: Supplier information, cost per unit (c_jk), and maximum supply
capacity (s_jk)
suppliers = ['Supplier1', 'Supplier2', 'Supplier3']
resources = ['Toilet_Paper', 'Liquid_Soap', 'Detergent', 'Cloths',
            'Toothpaste', 'Toothbrushes', 'Sanitary_Pads', 'Shampoo']

# Cost and maximum supply capacity for each supplier and resource
cost_per_unit = {
    ('Supplier1', 'Toilet_Paper'): (0.80, 150),
    ('Supplier1', 'Liquid_Soap'): (6.40, 25),
    ('Supplier1', 'Detergent'): (6.80, 20),
    ('Supplier1', 'Cloths'): (10.00, 10),
    ('Supplier1', 'Toothpaste'): (2.60, 50),
    ('Supplier1', 'Toothbrushes'): (0.80, 50),
    ('Supplier1', 'Sanitary_Pads'): (0.20, 150),
    ('Supplier1', 'Shampoo'): (2.30, 20),
    ('Supplier2', 'Toilet_Paper'): (0.95, 100),
    ('Supplier2', 'Liquid_Soap'): (3.98, 15),
    ('Supplier2', 'Detergent'): (4.60, 10),
    ('Supplier2', 'Cloths'): (11.00, 10),
    ('Supplier2', 'Toothpaste'): (3.00, 60),
    ('Supplier2', 'Toothbrushes'): (0.85, 60),
    ('Supplier2', 'Sanitary_Pads'): (0.18, 100),
    ('Supplier2', 'Shampoo'): (1.20, 20),
    ('Supplier3', 'Toilet_Paper'): (0.84, 70),
    ('Supplier3', 'Liquid_Soap'): (5.50, 30),
    ('Supplier3', 'Detergent'): (7.50, 15),
    ('Supplier3', 'Cloths'): (10.50, 15),
    ('Supplier3', 'Toothpaste'): (2.80, 30),
    ('Supplier3', 'Toothbrushes'): (0.82, 30),
    ('Supplier3', 'Sanitary_Pads'): (0.15, 100),
    ('Supplier3', 'Shampoo'): (3.00, 30)
}

# Minimum required quantities as a dataset for a shelter housing 20 people
for one month
minimum_quantities = {
    'Toilet_Paper': 200,
    'Liquid_Soap': 40,
    'Detergent': 30,
    'Cloths': 20,
    'Toothpaste': 100,
    'Toothbrushes': 100,
    'Sanitary_Pads': 300,
    'Shampoo': 40
}
```

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Correct Answer 2.

Let x_{jk} represent the number of units of resource j purchased from supplier k , where $j = 1, \dots, 8$ (resource types: Toilet Paper, Liquid Soap, Detergent, Cloths, Toothpaste, Toothbrushes, Sanitary Pads, Shampoo) and $k = 1, 2, 3$ (suppliers: Supplier 1, Supplier 2, Supplier 3).

Our objective is to minimize the cost of acquiring resources for the shelter, given in (2a).

Other potential objective is to maximize the supply of resources for the shelter, given in (1).

$$\text{Max } Z = \sum_{j=1}^8 \sum_{k=1}^3 (x_{jk}) \quad (1)$$

For each type of resource, the minimum required quantity must be met, shown in (2b). Budget for the month, \$2,000 should not be exceeded, given in (2c). The suppliers have limited resources and cannot deliver more than their upper limit, given in (2d). The non-negativity constraints ensure that all variables are positive or zero, as shown in (2e).

Linear Program:

$$\text{Minimize}_{x_{jk}} \quad \sum_{j=1}^8 \sum_{k=1}^3 (x_{jk} \cdot c_{jk}) \quad (2a)$$

$$\text{Subject to} \quad \sum_{k=1}^3 x_{jk} \geq m_j, \quad \forall j \in \{1, \dots, 8\} \quad (2b)$$

$$\sum_{j=1}^8 \sum_{k=1}^3 (x_{jk} \cdot c_{jk}) \leq 2000 \quad \forall k \in \{1, 2, 3\} \quad (2c)$$

$$x_{jk} \leq s_{jk} \quad \forall j \in \{1, \dots, 8\}, k \in \{1, 2, 3\} \quad (2d)$$

$$x_{jk} \geq 0 \quad \forall j \in \{1, \dots, 8\}, k \in \{1, 2, 3\} \quad (2e)$$

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```
# Create a new model
m = Model("resource_optimization")

# Decision Variables
x = {}
for r in resources:
    for s in suppliers:
        x[(r, s)] = m.addVar(name=f"x_{r}_{s}")

# Objective: Minimize cost
m.setObjective(quicksum(x[r, s] * cost_per_unit[s, r][0] for r in resources
                        for s in suppliers), GRB.MINIMIZE)

# Constraints
# Minimum requirements
for r in resources:
    m.addConstr(quicksum(x[r, s] for s in suppliers) >= minimum_quantities[
        r], name=f"min_req_{r}")

# Budget
m.addConstr(quicksum(x[r, s] * cost_per_unit[s, r][0] for r in resources
                    for s in suppliers) <= 2000, name=f"budget_{s}")

# Maximum supply limit per item from each supplier
for r in resources:
    for s in suppliers:
        m.addConstr(x[r, s] <= cost_per_unit[s, r][1], name=f"max_supply_{r}_{s}")

# Solve
m.optimize()

# Print solution
if m.status == GRB.OPTIMAL:
    for r in resources:
        for s in suppliers:
            if x[r, s].x > 0:
                print(f"Purchase_{x[r,s].x}units_of_{r}_from_{s}")
else:
    print("No feasible solution found")
```

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The optimal solution under the objective given in (2a) recommends shelter to follow this purchasing plan:

- Purchase 150.0 units of Toilet Paper from Supplier1
- Purchase 50.0 units of Toilet Paper from Supplier3
- Purchase 15.0 units of Liquid Soap from Supplier2
- Purchase 25.0 units of Liquid Soap from Supplier3
- Purchase 20.0 units of Detergent from Supplier1
- Purchase 10.0 units of Detergent from Supplier2
- Purchase 10.0 units of Cloths from Supplier1
- Purchase 10.0 units of Cloths from Supplier3
- Purchase 50.0 units of Toothpaste from Supplier1
- Purchase 20.0 units of Toothpaste from Supplier2
- Purchase 30.0 units of Toothpaste from Supplier3
- Purchase 50.0 units of Toothbrushes from Supplier1
- Purchase 20.0 units of Toothbrushes from Supplier2
- Purchase 30.0 units of Toothbrushes from Supplier3
- Purchase 100.0 units of Sanitary Pads from Supplier1
- Purchase 100.0 units of Sanitary Pads from Supplier2
- Purchase 100.0 units of Sanitary Pads from Supplier3
- Purchase 20.0 units of Shampoo from Supplier1
- Purchase 20.0 units of Shampoo from Supplier2

The total purchase amounts to \$1,224.80, (using (2a)) which represents approximately 61% of the monthly budget of \$2,000. Given this significant portion of the budget being allocated towards these supplies, careful consideration should be given to the inventory management strategy. Specifically, when it comes to shelter operations, it is generally more advantageous to overstock rather than under-stock essential items. Overstocking helps ensure that there are always enough supplies to meet the needs of the shelter, avoiding the potential risks and costs associated with running out of critical items. Running out could lead to disruptions in shelter operations, increased expenses due to emergency purchases, or even diminished quality of care. Additionally, considering the current inventory, it may be cost-effective in the long run to maintain a buffer stock, as the cost of not having these items readily available could outweigh the expense of purchasing more than immediately necessary.